asset-class-performance-fed-policy-cycles

January 28, 2026

1 Performance Of Various Asset Classes During Fed Policy Cycles

1.1 Python Imports

[1]:
```
# Standard Library
import datetime
import io
import os
import random
import sys
import warnings

from datetime import datetime, timedelta
from pathlib import Path

# Data Handling
import numpy as np
import pandas as pd

# Data Visualization
import matplotlib.dates as mdates
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
from matplotlib.ticker import FormatStrFormatter, FuncFormatter, MultipleLocator

# Data Sources
import yfinance as yf
import pandas_datareader.data as web

# Statistical Analysis
import statsmodels.api as sm

# Machine Learning
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

# Suppress warnings
```

```
warnings.filterwarnings("ignore")
```

1.2 Add Directories To Path

[2]:
```
# Add the source subdirectory to the system path to allow import config from␣
↪settings.py
current_directory = Path(os.getcwd())
website_base_directory = current_directory.parent.parent.parent
src_directory = website_base_directory / "src"
sys.path.append(str(src_directory)) if str(src_directory) not in sys.path else␣
↪None

# Import settings.py
from settings import config

# Add configured directories from config to path
SOURCE_DIR = config("SOURCE_DIR")
sys.path.append(str(Path(SOURCE_DIR))) if str(Path(SOURCE_DIR)) not in sys.path␣
↪else None

# Add other configured directories
BASE_DIR = config("BASE_DIR")
CONTENT_DIR = config("CONTENT_DIR")
POSTS_DIR = config("POSTS_DIR")
PAGES_DIR = config("PAGES_DIR")
PUBLIC_DIR = config("PUBLIC_DIR")
SOURCE_DIR = config("SOURCE_DIR")
DATA_DIR = config("DATA_DIR")
DATA_MANUAL_DIR = config("DATA_MANUAL_DIR")

# Print system path
for i, path in enumerate(sys.path):
    print(f"{i}: {path}")
```

```
0: /usr/lib/python313.zip
1: /usr/lib/python3.13
2: /usr/lib/python3.13/lib-dynload
3:
4: /home/jared/python-virtual-envs/general-venv-p313/lib/python3.13/site-
packages
5: /home/jared/python-virtual-envs/general-venv-p313/lib/python3.13/site-
packages/setuptools/_vendor
6:
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io_congo/src
```

1.3 Track Index Dependencies

[3]:
```
# Create file to track markdown dependencies
dep_file = Path("index_dep.txt")
dep_file.write_text("")
```

[3]: 0

1.4 Python Functions

[4]:
```
from calc_fed_cycle_asset_performance import calc_fed_cycle_asset_performance
from df_info import df_info
from df_info_markdown import df_info_markdown
from export_track_md_deps import export_track_md_deps
from load_data import load_data
from pandas_set_decimal_places import pandas_set_decimal_places
from plot_bar_returns_ffr_change import plot_bar_returns_ffr_change
from plot_timeseries import plot_timeseries
from plot_scatter_regression_ffr_vs_returns import␣
↪plot_scatter_regression_ffr_vs_returns
from sm_ols_summary_markdown import sm_ols_summary_markdown
from summary_stats import summary_stats
from yf_pull_data import yf_pull_data
```

1.5 Data Overview

[5]:
```
# Set timeframe
start_date = "2004-11-30"  # GLD inception (month end)
end_date = "2025-10-31"
```

1.5.1 Acquire & Plot Fed Funds Data

[6]:
```
# Set decimal places
pandas_set_decimal_places(4)

# Pull Effective Fed Funds Rate from FRED
fedfunds = web.DataReader("FEDFUNDS", "fred", start="1900-01-01", end=datetime.
↪today())
fedfunds["FEDFUNDS"] = fedfunds["FEDFUNDS"] / 100  # Convert to decimal

# Resample to monthly frequency and compute change in rate
fedfunds_monthly = fedfunds.resample("M").last()
fedfunds_monthly = fedfunds_monthly[(fedfunds_monthly.index >= pd.
↪to_datetime(start_date)) & (fedfunds_monthly.index <= pd.
↪to_datetime(end_date))]
fedfunds_monthly["FedFunds_Change"] = fedfunds_monthly["FEDFUNDS"].diff()
```

[7]:
```
df_info(fedfunds_monthly)
```

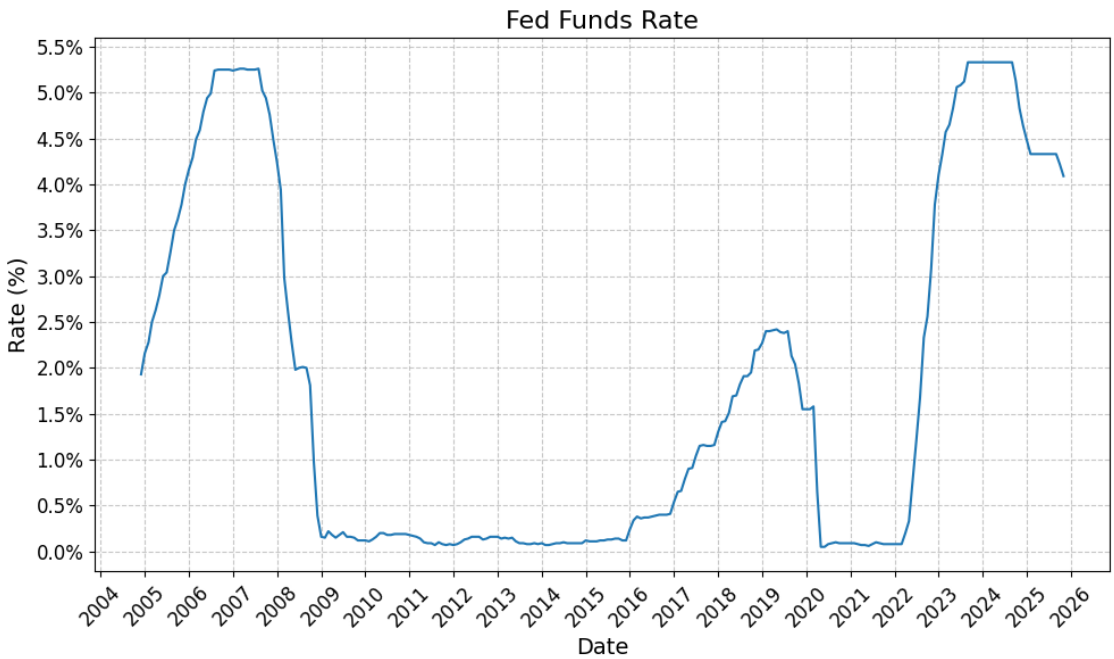
```
The columns, shape, and data types are:
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 252 entries, 2004-11-30 to 2025-10-31
Freq: ME
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   FEDFUNDS         252 non-null    float64
 1   FedFunds_Change  251 non-null    float64
dtypes: float64(2)
memory usage: 5.9 KB
None
The first 5 rows are:

            FEDFUNDS  FedFunds_Change
DATE
2004-11-30    0.0193              NaN
2004-12-31    0.0216           0.0023
2005-01-31    0.0228           0.0012
2005-02-28    0.0250           0.0022
2005-03-31    0.0263           0.0013

The last 5 rows are:

            FEDFUNDS  FedFunds_Change
DATE
2025-06-30    0.0433           0.0000
2025-07-31    0.0433           0.0000
2025-08-31    0.0433           0.0000
2025-09-30    0.0422          -0.0011
2025-10-31    0.0409          -0.0013
```

[8]:
```
# Copy this <!-- INSERT_01_Fed_Funds_Monthly_Rate_Change_HERE --> to index_temp.
↪md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="01_Fed_Funds_Monthly_Rate_Change.md",
    content=df_info_markdown(df=fedfunds_monthly, decimal_places=4),
    output_type="markdown",
)
```

```
Exported and tracked: 01_Fed_Funds_Monthly_Rate_Change.md
```

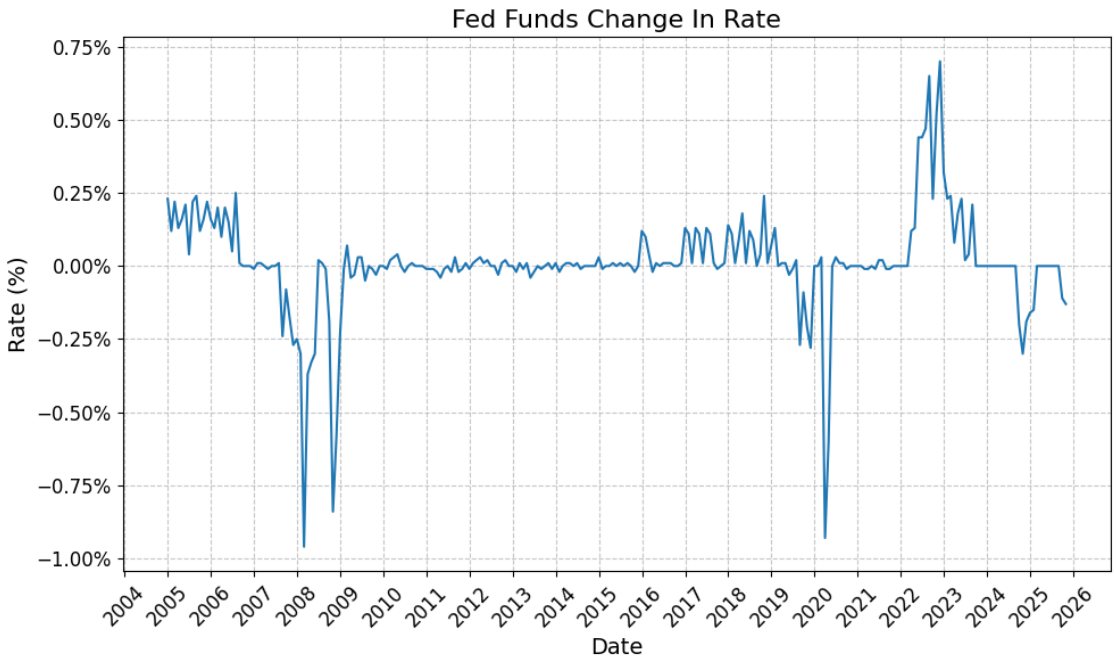
[9]:
```
plot_timeseries(
    price_df=fedfunds_monthly,
    plot_start_date=start_date,
    plot_end_date=end_date,
```

```
    plot_columns=["FEDFUNDS"],
    title="Fed Funds Rate",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Rate (%)",
    y_format="Percentage",
    y_format_decimal_places=1,
    y_tick_spacing=0.005,
    grid=True,
    legend=False,
    export_plot=True,
    plot_file_name="01_Fed_Funds_Rate",
)
```



[10]:
```
plot_timeseries(
    price_df=fedfunds_monthly,
    plot_start_date=start_date,
    plot_end_date=end_date,
    plot_columns=["FedFunds_Change"],
    title="Fed Funds Change In Rate",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Rate (%)",
```

```
    y_format="Percentage",
    y_format_decimal_places=2,
    y_tick_spacing=0.0025,
    grid=True,
    legend=False,
    export_plot=True,
    plot_file_name="01_Fed_Funds_Change_In_Rate",
)
```



1.5.2 Define Fed Policy Cycles

[11]:
```
# # Define manually specified Fed policy cycles
# fed_cycles = [
#     # ("2002-01-01", "2003-07-01"),
#     # ("2003-07-01", "2004-06-01"),
#     # ("2004-06-01", "2006-07-01"),
#     ("2004-11-01", "2006-07-01"),
#     ("2006-07-01", "2007-07-01"),
#     ("2007-07-01", "2008-12-01"),
#     ("2008-12-01", "2015-11-01"),
#     ("2015-11-01", "2019-01-01"),
#     ("2019-01-01", "2019-07-01"),
#     ("2019-07-01", "2020-04-01"),
#     ("2020-04-01", "2022-02-01"),
#     ("2022-02-01", "2023-08-01"),
```

```
#     ("2023-08-01", "2024-08-01"),
#     ("2024-08-01", datetime.today().strftime('%Y-%m-%d')),
# ]

# # Optional: assign a name to each cycle
# cycle_labels = [f"Cycle {i+1}" for i in range(len(fed_cycles))]
```

[12]:
```
# Define manually specified Fed policy cycles
fed_cycles = [
    ("2004-11-01", "2006-07-01"),
    ("2006-07-01", "2007-07-01"),
    ("2007-07-01", "2008-12-01"),
    ("2008-12-01", "2015-11-01"),
    ("2015-11-01", "2019-01-01"),
    ("2019-01-01", "2019-07-01"),
    ("2019-07-01", "2020-04-01"),
    ("2020-04-01", "2022-02-01"),
    ("2022-02-01", "2023-08-01"),
    ("2023-08-01", "2024-08-01"),
    ("2024-08-01", datetime.today().strftime('%Y-%m-%d')),
]

# Optional: assign a name to each cycle
cycle_labels = [f"Cycle {i+1}" for i in range(len(fed_cycles))]
```

[13]:
```
# Set decimal places
pandas_set_decimal_places(4)

#  Calc changes by fed cycle defined above
fed_changes = []

for (start, end) in fed_cycles:
    start = pd.to_datetime(start)
    end = pd.to_datetime(end)

    try:
        rate_start = fedfunds.loc[start, "FEDFUNDS"]
    except KeyError:
        rate_start = fedfunds.loc[:start].iloc[-1]["FEDFUNDS"]

    try:
        rate_end = fedfunds.loc[end, "FEDFUNDS"]
    except KeyError:
        rate_end = fedfunds.loc[:end].iloc[-1]["FEDFUNDS"]

    change = rate_end - rate_start
    fed_changes.append(change)
```

```
fed_changes_df = pd.DataFrame({
    "Cycle": cycle_labels,
    "FedFunds_Change": fed_changes
})

fed_changes_df
```

[13]:

	Cycle	FedFunds_Change
0	Cycle 1	0.0331
1	Cycle 2	0.0002
2	Cycle 3	-0.0510
3	Cycle 4	-0.0004
4	Cycle 5	0.0228
5	Cycle 6	0.0000
6	Cycle 7	-0.0235
7	Cycle 8	0.0003
8	Cycle 9	0.0525
9	Cycle 10	0.0000
10	Cycle 11	-0.0161

[14]:
```
# Copy this <!-- INSERT_01_Fed_Funds_Cycle_Change_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="01_Fed_Funds_Cycle_Change.md",
    content=fed_changes_df.to_markdown(floatfmt=".4f"),
    output_type="markdown",
)
```

```
Exported and tracked: 01_Fed_Funds_Cycle_Change.md
```

1.6 Return Performance By Fed Policy Cycle

1.6.1 Stocks (SPY)

[15]:
```
# Set decimal places
pandas_set_decimal_places(2)

yf_pull_data(
    base_directory=DATA_DIR,
    ticker="SPY",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    excel_export=True,
    pickle_export=True,
    output_confirmation=True,
)
```

```
[*********************100%***********************]  1 of 1 completed
```

```
The first and last date of data for SPY is:

            Close  High   Low  Open   Volume
Date
1993-01-29  24.24 24.26 24.14 24.26  1003200

            Close   High    Low   Open    Volume
Date
2026-01-27 695.49 696.53 693.57 694.18  55506100

Yahoo Finance data complete for SPY
--------------------
```

[15]:

```
             Close   High    Low   Open     Volume
Date
1993-01-29   24.24  24.26  24.14  24.26    1003200
1993-02-01   24.41  24.41  24.26  24.26     480500
1993-02-02   24.47  24.48  24.34  24.40     201300
1993-02-03   24.72  24.74  24.48  24.50     529400
1993-02-04   24.83  24.88  24.53  24.81     531500
…            …      …      …      …         …
2026-01-21  685.40 688.74 678.13 679.65  127844500
2026-01-22  688.98 691.13 686.92 689.85   77112200
2026-01-23  689.23 690.96 687.16 688.15   63059600
2026-01-26  692.73 694.13 689.92 690.49   60473800
2026-01-27  695.49 696.53 693.57 694.18   55506100

[8305 rows x 5 columns]
```

[16]:

```
spy = load_data(
    base_directory=DATA_DIR,
    ticker="SPY",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    timeframe="Daily",
    file_format="pickle",
)

# Filter SPY to date range
spy = spy[(spy.index >= pd.to_datetime(start_date)) & (spy.index <= pd.
 ↪to_datetime(end_date))]

# Resample to monthly frequency
spy_monthly = spy.resample("M").last()
spy_monthly["Monthly_Return"] = spy_monthly["Close"].pct_change()
```

[17]:

```
df_info(spy_monthly)
```

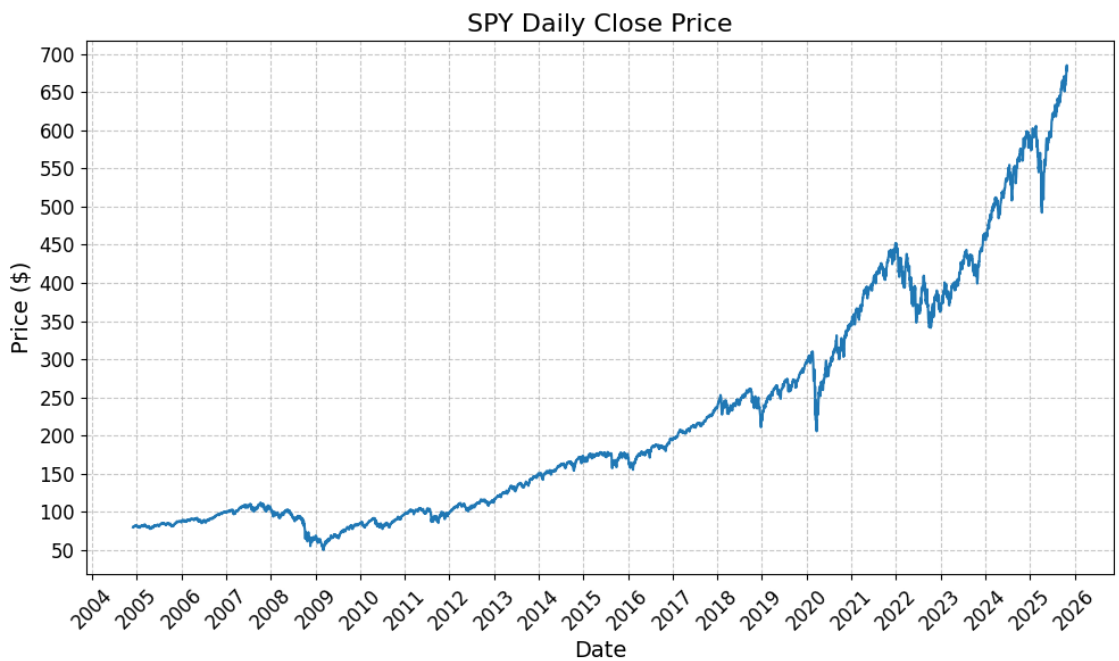
```
The columns, shape, and data types are:
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 252 entries, 2004-11-30 to 2025-10-31
Freq: ME
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Close           252 non-null    float64
 1   High            252 non-null    float64
 2   Low             252 non-null    float64
 3   Open            252 non-null    float64
 4   Volume          252 non-null    int64
 5   Monthly_Return  251 non-null    float64
dtypes: float64(5), int64(1)
memory usage: 13.8 KB
None
The first 5 rows are:

            Close  High   Low  Open    Volume  Monthly_Return
Date
2004-11-30  79.60 79.83 79.43 79.67  53685200             NaN
2004-12-31  81.99 82.53 81.95 82.28  28648800            0.03
2005-01-31  80.15 80.22 79.85 80.01  52532700           -0.02
2005-02-28  81.83 82.28 81.43 82.18  69381300            0.02
2005-03-31  80.33 80.67 80.27 80.49  64575400           -0.02

The last 5 rows are:

            Close   High    Low   Open     Volume  Monthly_Return
Date
2025-06-30 614.33 615.69 611.53 613.86   92502500            0.05
2025-07-31 628.48 636.20 627.17 635.81  103385200            0.02
2025-08-31 641.37 644.15 639.47 643.78   74522200            0.02
2025-09-30 664.22 664.69 659.66 660.98   86288000            0.04
2025-10-31 680.05 683.06 677.24 683.02   87164100            0.02
```

[18]:
```python
# Copy this <!-- INSERT_02_SPY_Monthly_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="02_SPY_Monthly.md",
    content=df_info_markdown(df=spy_monthly, decimal_places=2),
    output_type="markdown",
)
```

```
Exported and tracked: 02_SPY_Monthly.md
```

[19]:
```python
plot_timeseries(
    price_df=spy,
    plot_start_date=start_date,
```

```
    plot_end_date=end_date,
    plot_columns=["Close"],
    title="SPY Daily Close Price",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=50,
    grid=True,
    legend=False,
    export_plot=True,
    plot_file_name="02_SPY_Price",
)
```



[20]:
```
spy_cycle_df = calc_fed_cycle_asset_performance(
    fed_cycles=fed_cycles,
    cycle_labels=cycle_labels,
    fed_changes=fed_changes,
    monthly_df=spy_monthly,
)

spy_cycle_df
```

[20]:

	Cycle	Start	End	Months	CumulativeReturn	\
0	Cycle 1	2004-11-01	2006-07-01	20	0.11	
1	Cycle 2	2006-07-01	2007-07-01	12	0.20	
2	Cycle 3	2007-07-01	2008-12-01	17	-0.39	
3	Cycle 4	2008-12-01	2015-11-01	83	1.67	
4	Cycle 5	2015-11-01	2019-01-01	38	0.28	
5	Cycle 6	2019-01-01	2019-07-01	6	0.18	
6	Cycle 7	2019-07-01	2020-04-01	9	-0.11	
7	Cycle 8	2020-04-01	2022-02-01	22	0.79	
8	Cycle 9	2022-02-01	2023-08-01	18	0.04	
9	Cycle 10	2023-08-01	2024-08-01	12	0.22	
10	Cycle 11	2024-08-01	2026-01-28	15	0.26	

	CumulativeReturnPct	AverageMonthlyReturn	AverageMonthlyReturnPct	\
0	11.32	0.01	0.59	
1	20.36	0.02	1.57	
2	-38.55	-0.03	-2.67	
3	167.34	0.01	1.28	
4	28.30	0.01	0.70	
5	18.33	0.03	2.95	
6	-10.67	-0.01	-1.10	
7	79.13	0.03	2.78	
8	4.18	0.00	0.40	
9	22.00	0.02	1.75	
10	25.72	0.02	1.59	

	AnnualizedReturn	AnnualizedReturnPct	Volatility	FedFundsChange	\
0	0.07	6.64	0.08	0.03	
1	0.20	20.36	0.07	0.00	
2	-0.29	-29.09	0.19	-0.05	
3	0.15	15.28	0.15	-0.00	
4	0.08	8.19	0.11	0.02	
5	0.40	40.01	0.18	0.00	
6	-0.14	-13.96	0.19	-0.02	
7	0.37	37.43	0.16	0.00	
8	0.03	2.77	0.21	0.05	
9	0.22	22.00	0.15	0.00	
10	0.20	20.09	0.11	-0.02	

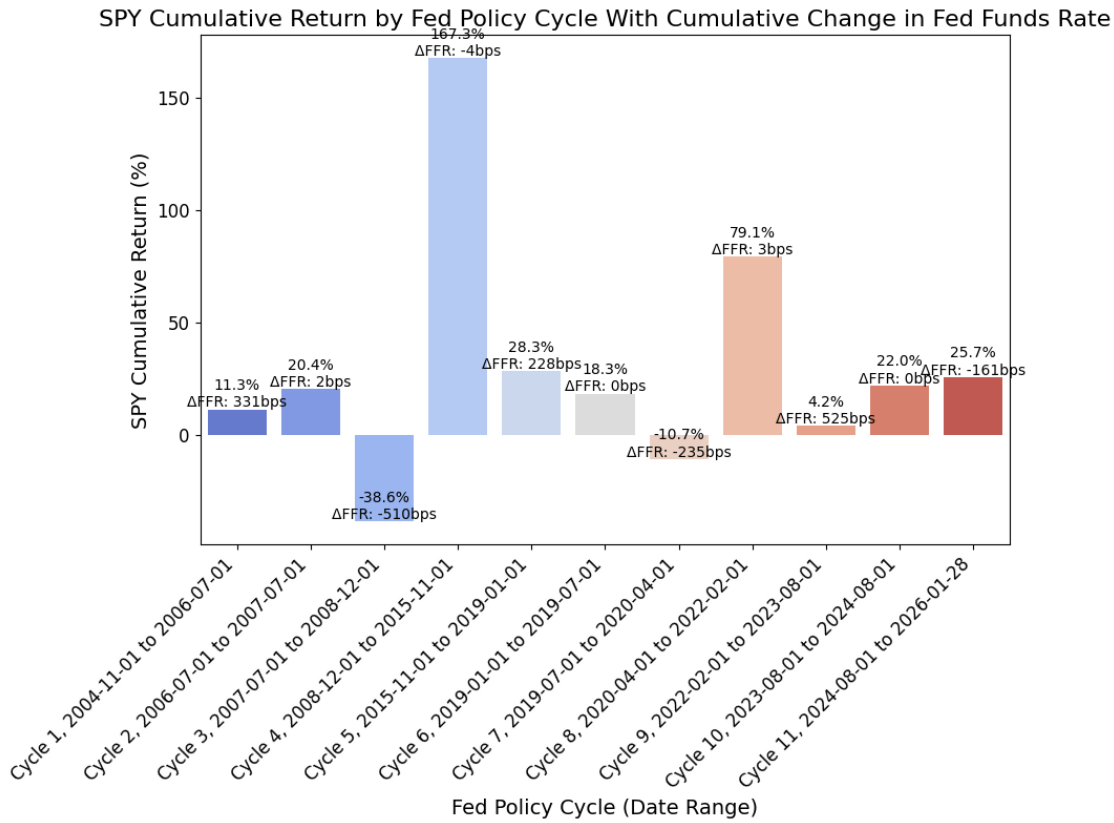
	FedFundsChange_bps	FFR_AnnualizedChange	FFR_AnnualizedChange_bps	\
0	331.00	0.02	198.60	
1	2.00	0.00	2.00	
2	-510.00	-0.04	-360.00	
3	-4.00	-0.00	-0.58	
4	228.00	0.01	72.00	
5	0.00	0.00	0.00	
6	-235.00	-0.03	-313.33	

```
7                 3.00                  0.00                      1.64
8               525.00                  0.03                    350.00
9                 0.00                  0.00                      0.00
10             -161.00                 -0.01                   -128.80

                                 Label
0    Cycle 1, 2004-11-01 to 2006-07-01
1    Cycle 2, 2006-07-01 to 2007-07-01
2    Cycle 3, 2007-07-01 to 2008-12-01
3    Cycle 4, 2008-12-01 to 2015-11-01
4    Cycle 5, 2015-11-01 to 2019-01-01
5    Cycle 6, 2019-01-01 to 2019-07-01
6    Cycle 7, 2019-07-01 to 2020-04-01
7    Cycle 8, 2020-04-01 to 2022-02-01
8    Cycle 9, 2022-02-01 to 2023-08-01
9   Cycle 10, 2023-08-01 to 2024-08-01
10  Cycle 11, 2024-08-01 to 2026-01-28
```

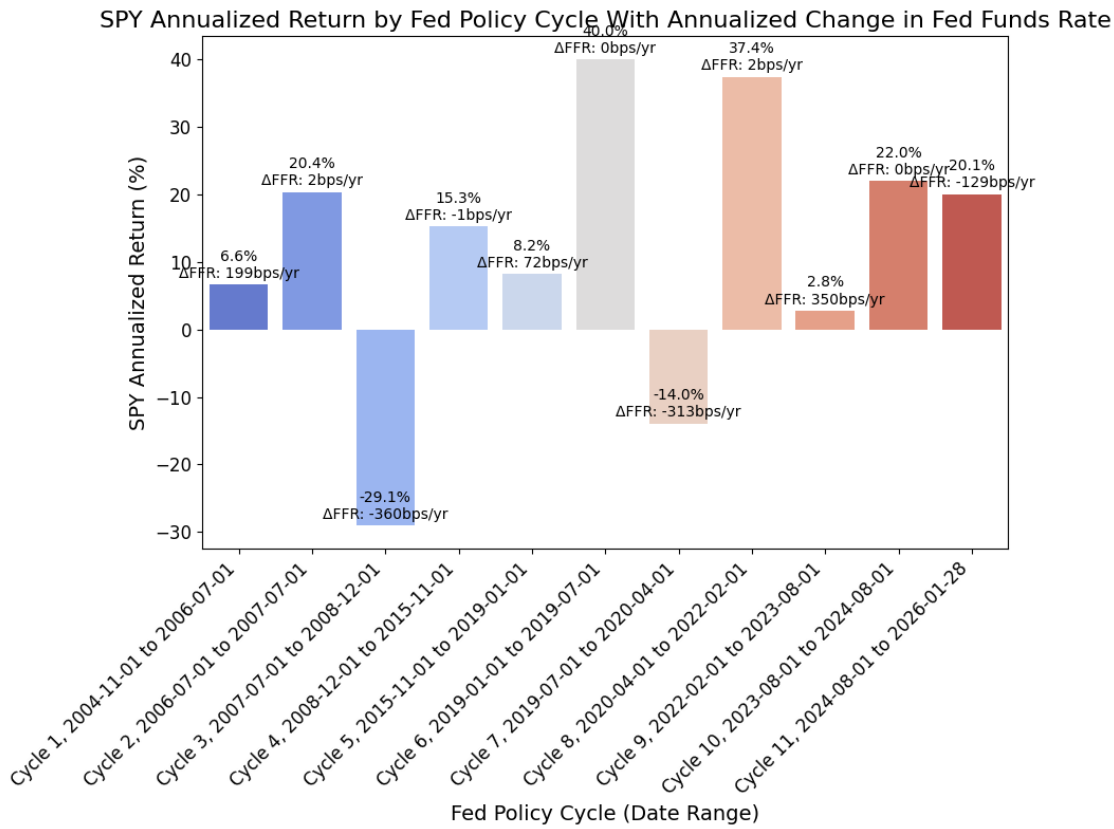
[21]:
```
# Copy this <!-- INSERT_02_SPY_Cycle_DF_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="02_SPY_Cycle_DF.md",
    content=spy_cycle_df.to_markdown(floatfmt=".2f"),
    output_type="markdown",
)
```

```
Exported and tracked: 02_SPY_Cycle_DF.md
```

[22]:
```
plot_bar_returns_ffr_change(
    cycle_df=spy_cycle_df,
    asset_label="SPY",
    annualized_or_cumulative="Cumulative",
    index_num="02",
)
```



[23]:
```
plot_bar_returns_ffr_change(
    cycle_df=spy_cycle_df,
    asset_label="SPY",
    annualized_or_cumulative="Annualized",
    index_num="02",
)
```



[24]:
```
df = spy_cycle_df

#####################################
### Don't modify below this line ###
#####################################

# Run OLS regression with statsmodels
X = df["FFR_AnnualizedChange_bps"]
y = df["AnnualizedReturnPct"]
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())
print(f"Intercept: {model.params[0]}, Slope: {model.params[1]}")  # Intercept␣
↪and slope

# Calc X and Y values for regression line
X_vals = np.linspace(X.min(), X.max(), 100)
Y_vals = model.params[0] + model.params[1] * X_vals
```

```
                            OLS Regression Results
==============================================================================
```

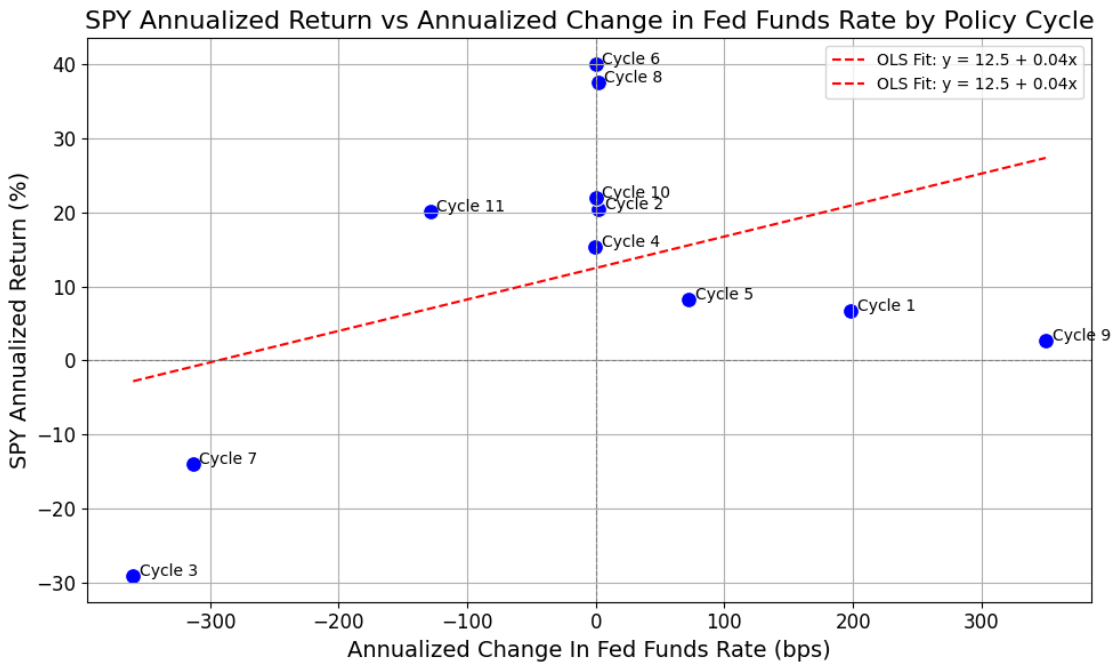
```
Dep. Variable:     AnnualizedReturnPct   R-squared:                       0.176
Model:                             OLS   Adj. R-squared:                  0.085
Method:                  Least Squares   F-statistic:                     1.928
Date:                 Wed, 28 Jan 2026   Prob (F-statistic):              0.198
Time:                         19:49:52   Log-Likelihood:                -47.196
No. Observations:                   11   AIC:                             98.39
Df Residuals:                        9   BIC:                             99.19
Df Model:                            1
Covariance Type:             nonrobust
==============================================================================
============
                              coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
------------
const                       12.4815      5.909      2.112      0.064      -0.886
25.849
FFR_AnnualizedChange_bps     0.0424      0.031      1.389      0.198      -0.027
0.112
==============================================================================
Omnibus:                        1.103   Durbin-Watson:                   3.070
Prob(Omnibus):                  0.576   Jarque-Bera (JB):                0.674
Skew:                           0.021   Prob(JB):                        0.714
Kurtosis:                       1.788   Cond. No.                         194.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
Intercept: 12.481505626003901, Slope: 0.042426203704362016
```

[25]:
```
# Copy this <!-- INSERT_02_SPY_Annualized_Regression_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="02_SPY_Annualized_Regression.md",
    content=sm_ols_summary_markdown(result=model,␣
 ↪file_path="02_SPY_Annualized_Regression.md"),
    output_type="text",
)
```

```
Exported and tracked: 02_SPY_Annualized_Regression.md
```

[26]:
```
plot_scatter_regression_ffr_vs_returns(
    cycle_df=spy_cycle_df,
    asset_label="SPY",
    index_num="02",
    x_vals=X_vals,
    y_vals=Y_vals,
```

```
    intercept=model.params[0],
    slope=model.params[1],
)
```



1.6.2 Bonds (TLT)

[27]:
```
# Set decimal places
pandas_set_decimal_places(2)

yf_pull_data(
    base_directory=DATA_DIR,
    ticker="TLT",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    excel_export=True,
    pickle_export=True,
    output_confirmation=True,
)
```

```
[*********************100%***********************]  1 of 1 completed


The first and last date of data for TLT is:

            Close  High   Low  Open  Volume
Date
```

```
2002-07-30  36.65 36.82 36.65 36.75    6100

            Close  High   Low  Open    Volume
Date
2026-01-27  87.80 88.25 87.68 88.11  35180100

Yahoo Finance data complete for TLT
--------------------
```

[27]:

```
             Close  High   Low  Open    Volume
Date
2002-07-30  36.65 36.82 36.65 36.75      6100
2002-07-31  37.10 37.22 36.82 36.84     29400
2002-08-01  37.31 37.32 37.11 37.11     25000
2002-08-02  37.70 37.81 37.26 37.39     52800
2002-08-05  37.86 37.96 37.70 37.78     61100
…           …    …    …    …        …
2026-01-21  87.31 87.48 86.62 86.80  51220100
2026-01-22  87.69 87.76 87.14 87.27  42419300
2026-01-23  87.93 88.03 87.50 87.83  35968800
2026-01-26  88.35 88.49 88.20 88.40  38882400
2026-01-27  87.80 88.25 87.68 88.11  35180100

[5912 rows x 5 columns]
```

[28]:

```
tlt = load_data(
    base_directory=DATA_DIR,
    ticker="TLT",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    timeframe="Daily",
    file_format="pickle",
)

# Filter TLT to date range
tlt = tlt[(tlt.index >= pd.to_datetime(start_date)) & (tlt.index <= pd.
 ↪to_datetime(end_date))]

# Resample to monthly frequency
tlt_monthly = tlt.resample("M").last()
tlt_monthly["Monthly_Return"] = tlt_monthly["Close"].pct_change()
```

[29]:

```
df_info(tlt_monthly)
```

```
The columns, shape, and data types are:
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 252 entries, 2004-11-30 to 2025-10-31
Freq: ME
Data columns (total 6 columns):
```

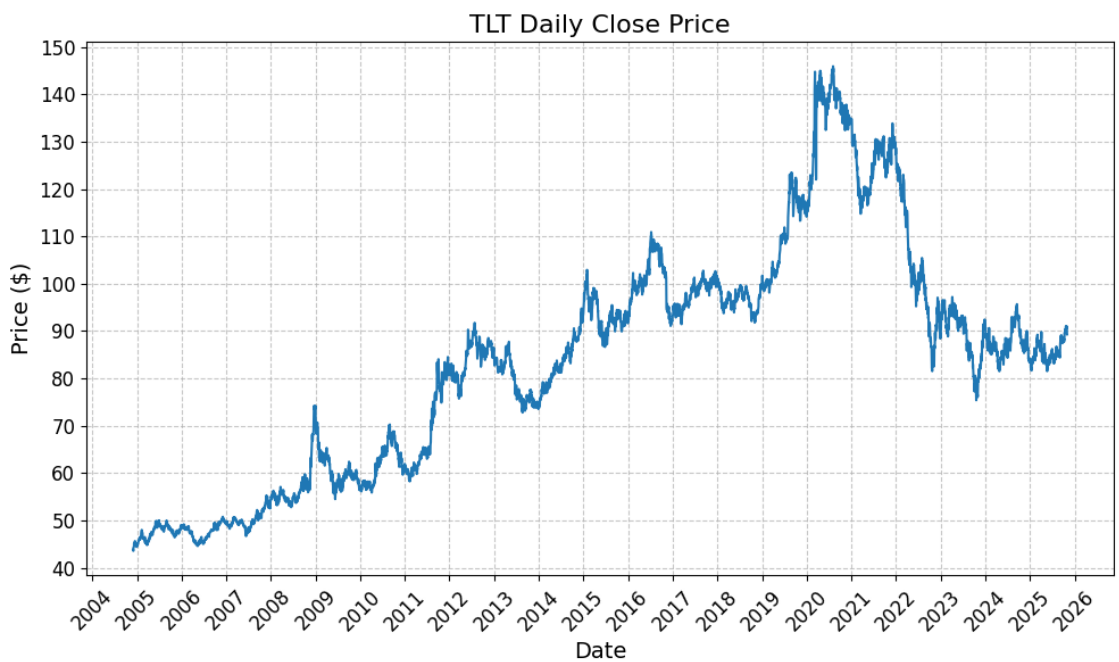
```
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Close           252 non-null    float64
 1   High            252 non-null    float64
 2   Low             252 non-null    float64
 3   Open            252 non-null    float64
 4   Volume          252 non-null    int64
 5   Monthly_Return  251 non-null    float64
dtypes: float64(5), int64(1)
memory usage: 13.8 KB
None
The first 5 rows are:

            Close  High   Low  Open   Volume  Monthly_Return
Date
2004-11-30  43.80 43.91 43.64 43.80  1754500             NaN
2004-12-31  44.96 45.01 44.84 44.87  1056400            0.03
2005-01-31  46.57 46.59 46.35 46.37  1313900            0.04
2005-02-28  45.88 46.43 45.82 46.43  2797300           -0.01
2005-03-31  45.67 45.70 45.43 45.61  2410900           -0.00

The last 5 rows are:

            Close  High   Low  Open    Volume  Monthly_Return
Date
2025-06-30  86.00 86.18 85.37 85.62  53695200            0.03
2025-07-31  85.02 85.50 84.94 85.22  49814100           -0.01
2025-08-31  85.03 85.28 84.87 85.18  41686400            0.00
2025-09-30  88.08 88.74 87.92 88.37  38584000            0.04
2025-10-31  89.30 89.66 89.21 89.56  38247300            0.01
```

[30]:
```
# Copy this <!-- INSERT_03_TLT_Monthly_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="03_TLT_Monthly.md",
    content=df_info_markdown(df=tlt_monthly, decimal_places=2),
    output_type="markdown",
)
```

```
Exported and tracked: 03_TLT_Monthly.md
```

[31]:
```
plot_timeseries(
    price_df=tlt,
    plot_start_date=start_date,
    plot_end_date=end_date,
    plot_columns=["Close"],
    title="TLT Daily Close Price",
    x_label="Date",
    x_format="Year",
```

```
    x_tick_rotation=45,
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=10,
    grid=True,
    legend=False,
    export_plot=True,
    plot_file_name="03_TLT_Price",
)
```



[32]:
```
tlt_cycle_df = calc_fed_cycle_asset_performance(
    fed_cycles=fed_cycles,
    cycle_labels=cycle_labels,
    fed_changes=fed_changes,
    monthly_df=tlt_monthly,
)

tlt_cycle_df
```

[32]:
```
       Cycle       Start         End  Months  CumulativeReturn  \
0    Cycle 1  2004-11-01  2006-07-01      20              0.04
1    Cycle 2  2006-07-01  2007-07-01      12              0.06
2    Cycle 3  2007-07-01  2008-12-01      17              0.32
3    Cycle 4  2008-12-01  2015-11-01      83              0.46
```

4	Cycle 5	2015-11-01	2019-01-01	38	0.07
5	Cycle 6	2019-01-01	2019-07-01	6	0.10
6	Cycle 7	2019-07-01	2020-04-01	9	0.26
7	Cycle 8	2020-04-01	2022-02-01	22	-0.11
8	Cycle 9	2022-02-01	2023-08-01	18	-0.27
9	Cycle 10	2023-08-01	2024-08-01	12	-0.02
10	Cycle 11	2024-08-01	2026-01-28	15	0.00

	CumulativeReturnPct	AverageMonthlyReturn	AverageMonthlyReturnPct	\
0	4.23	0.00	0.25	
1	5.76	0.00	0.49	
2	32.42	0.02	1.73	
3	45.67	0.01	0.55	
4	7.42	0.00	0.23	
5	10.48	0.02	1.73	
6	26.18	0.03	2.73	
7	-11.33	-0.00	-0.50	
8	-26.96	-0.02	-1.62	
9	-1.52	0.00	0.02	
10	0.42	0.00	0.08	

	AnnualizedReturn	AnnualizedReturnPct	Volatility	FedFundsChange	\
0	0.03	2.51	0.09	0.03	
1	0.06	5.76	0.07	0.00	
2	0.22	21.92	0.14	-0.05	
3	0.06	5.59	0.15	-0.00	
4	0.02	2.29	0.10	0.02	
5	0.22	22.05	0.13	0.00	
6	0.36	36.34	0.18	-0.02	
7	-0.06	-6.35	0.11	0.00	
8	-0.19	-18.90	0.17	0.05	
9	-0.02	-1.52	0.20	0.00	
10	0.00	0.33	0.11	-0.02	

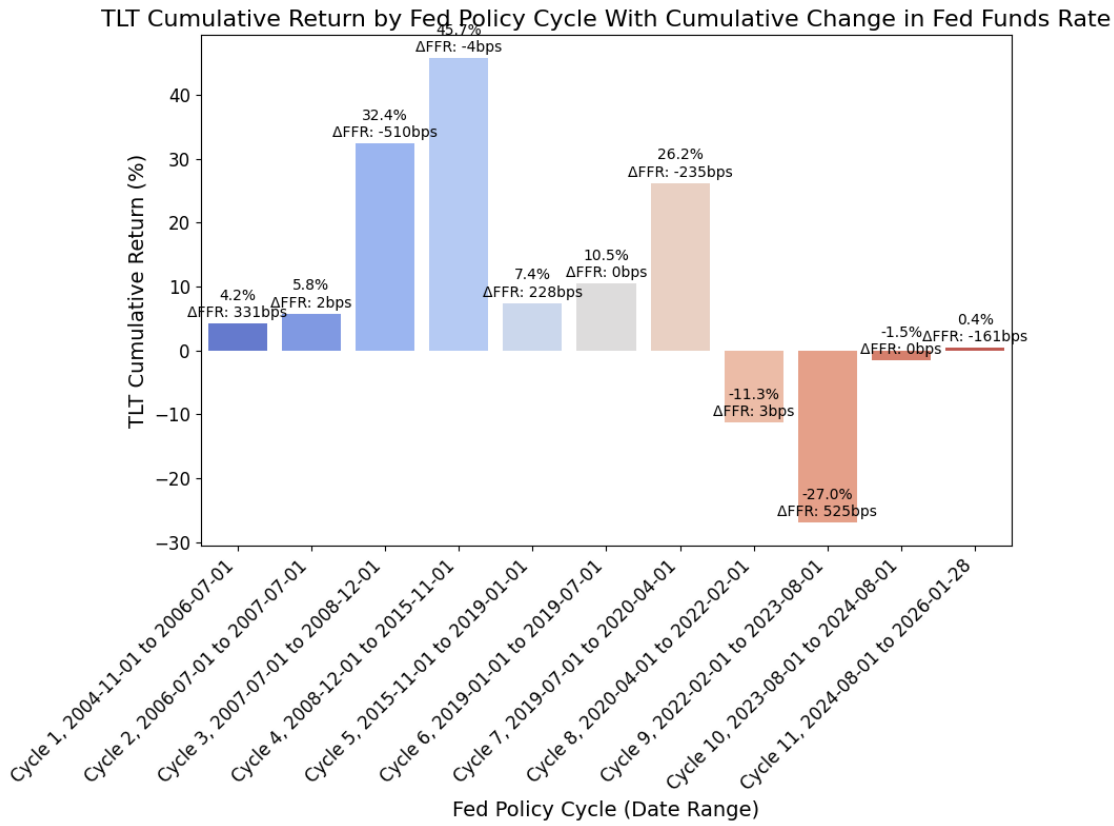
	FedFundsChange_bps	FFR_AnnualizedChange	FFR_AnnualizedChange_bps	\
0	331.00	0.02	198.60	
1	2.00	0.00	2.00	
2	-510.00	-0.04	-360.00	
3	-4.00	-0.00	-0.58	
4	228.00	0.01	72.00	
5	0.00	0.00	0.00	
6	-235.00	-0.03	-313.33	
7	3.00	0.00	1.64	
8	525.00	0.03	350.00	
9	0.00	0.00	0.00	
10	-161.00	-0.01	-128.80	

```
                                  Label
0    Cycle 1, 2004-11-01 to 2006-07-01
1    Cycle 2, 2006-07-01 to 2007-07-01
2    Cycle 3, 2007-07-01 to 2008-12-01
3    Cycle 4, 2008-12-01 to 2015-11-01
4    Cycle 5, 2015-11-01 to 2019-01-01
5    Cycle 6, 2019-01-01 to 2019-07-01
6    Cycle 7, 2019-07-01 to 2020-04-01
7    Cycle 8, 2020-04-01 to 2022-02-01
8    Cycle 9, 2022-02-01 to 2023-08-01
9   Cycle 10, 2023-08-01 to 2024-08-01
10  Cycle 11, 2024-08-01 to 2026-01-28
```

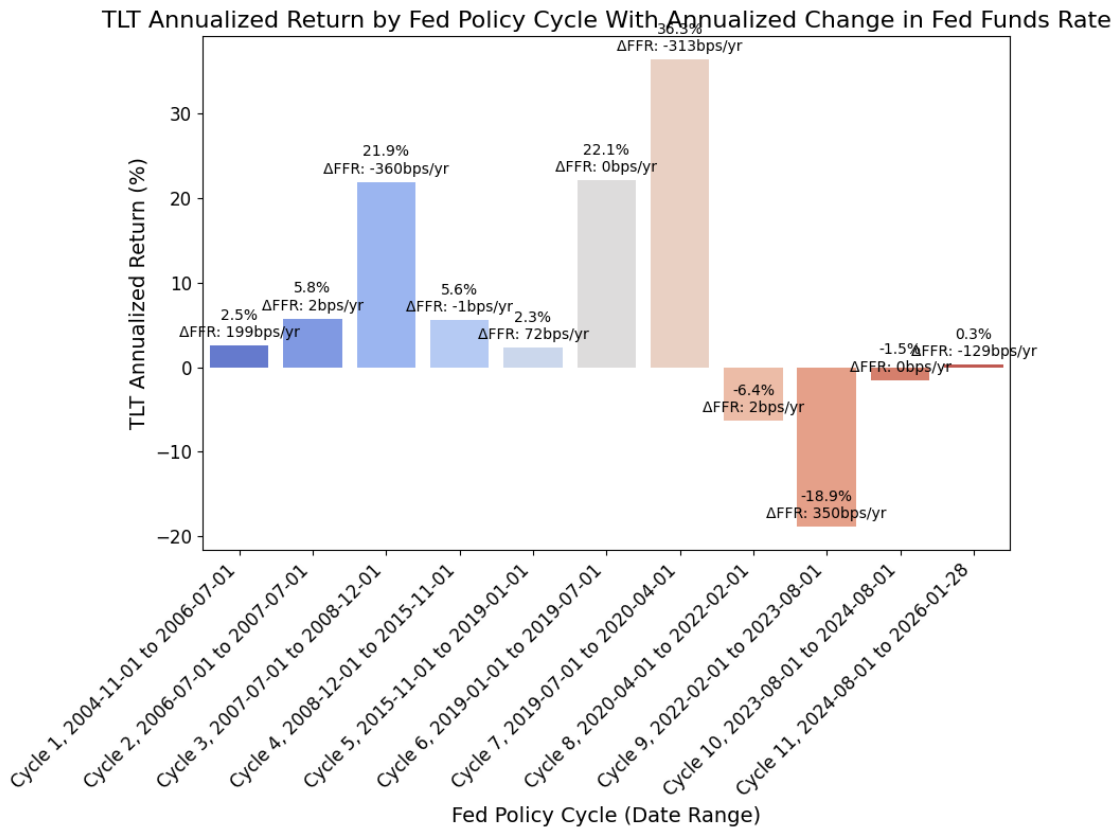
[33]:
```
# Copy this <!-- INSERT_03_TLT_Cycle_DF_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="03_TLT_Cycle_DF.md",
    content=tlt_cycle_df.to_markdown(floatfmt=".2f"),
    output_type="markdown",
)
```

```
Exported and tracked: 03_TLT_Cycle_DF.md
```

[34]:
```
plot_bar_returns_ffr_change(
    cycle_df=tlt_cycle_df,
    asset_label="TLT",
    annualized_or_cumulative="Cumulative",
    index_num="03",
)
```



[35]:
```
plot_bar_returns_ffr_change(
    cycle_df=tlt_cycle_df,
    asset_label="TLT",
    annualized_or_cumulative="Annualized",
    index_num="03",
)
```



[36]:
```
df = tlt_cycle_df

#####################################
### Don't modify below this line ###
#####################################

# Run OLS regression with statsmodels
X = df["FFR_AnnualizedChange_bps"]
y = df["AnnualizedReturnPct"]
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())
print(f"Intercept: {model.params[0]}, Slope: {model.params[1]}")  # Intercept␣
↪and slope

# Calc X and Y values for regression line
X_vals = np.linspace(X.min(), X.max(), 100)
Y_vals = model.params[0] + model.params[1] * X_vals
```

```
                            OLS Regression Results
==============================================================================
```

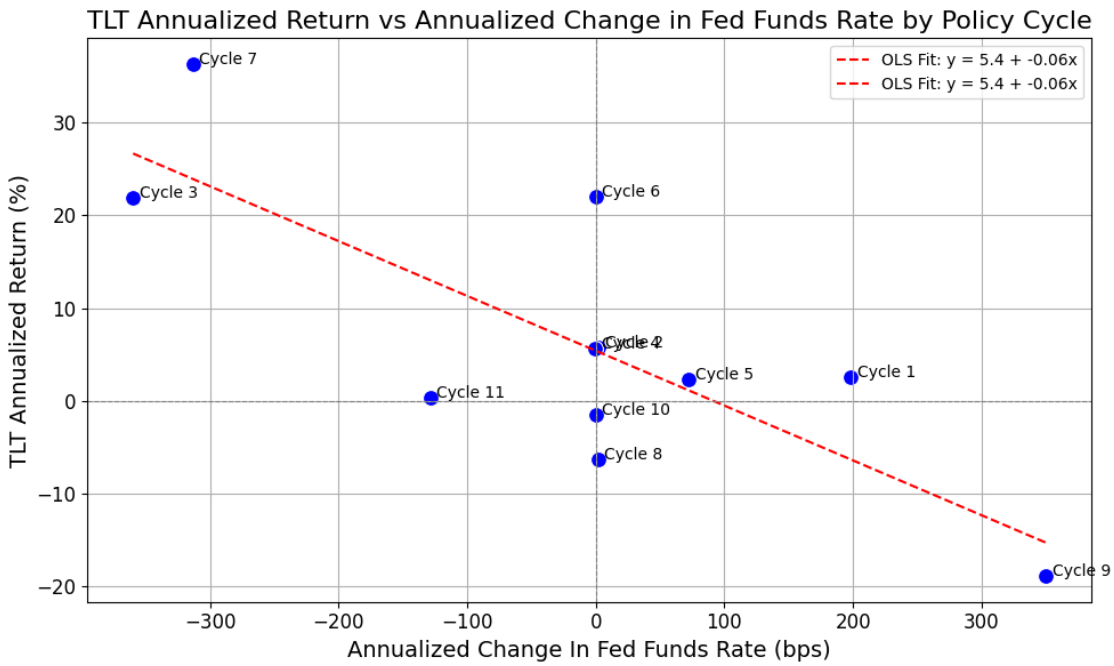
```
Dep. Variable:     AnnualizedReturnPct   R-squared:                       0.615
Model:                             OLS   Adj. R-squared:                  0.573
Method:                  Least Squares   F-statistic:                     14.39
Date:                 Wed, 28 Jan 2026   Prob (F-statistic):            0.00426
Time:                         19:50:00   Log-Likelihood:                -39.782
No. Observations:                   11   AIC:                             83.56
Df Residuals:                        9   BIC:                             84.36
Df Model:                            1
Covariance Type:             nonrobust
==============================================================================
============
                              coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
------------
const                        5.4077      3.012      1.796      0.106      -1.405
12.221
FFR_AnnualizedChange_bps    -0.0591      0.016     -3.794      0.004      -0.094
-0.024
==============================================================================
Omnibus:                        0.635   Durbin-Watson:                   1.199
Prob(Omnibus):                  0.728   Jarque-Bera (JB):                0.621
Skew:                           0.387   Prob(JB):                        0.733
Kurtosis:                       2.131   Cond. No.                         194.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
Intercept: 5.407704970962893, Slope: -0.059076942528379625
```

[37]:
```
# Copy this <!-- INSERT_03_TLT_Annualized_Regression_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="03_TLT_Annualized_Regression.md",
    content=sm_ols_summary_markdown(result=model,␣
 ↪file_path="03_TLT_Annualized_Regression.md"),
    output_type="text",
)
```

```
Exported and tracked: 03_TLT_Annualized_Regression.md
```

[38]:
```
plot_scatter_regression_ffr_vs_returns(
    cycle_df=tlt_cycle_df,
    asset_label="TLT",
    index_num="03",
    x_vals=X_vals,
    y_vals=Y_vals,
```

```
    intercept=model.params[0],
    slope=model.params[1],
)
```



1.6.3 Gold (GLD)

[39]:
```
# Set decimal places
pandas_set_decimal_places(2)

yf_pull_data(
    base_directory=DATA_DIR,
    ticker="GLD",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    excel_export=True,
    pickle_export=True,
    output_confirmation=True,
)
```

```
[*********************100%***********************]  1 of 1 completed


The first and last date of data for GLD is:

            Close  High   Low  Open   Volume
Date
```

```
2004-11-18  44.38 44.49 44.07 44.43  5992000

            Close   High    Low   Open    Volume
Date
2026-01-27 476.10 476.49 463.95 465.07  28386200

Yahoo Finance data complete for GLD
--------------------
```

[39]:

```
             Close   High    Low   Open    Volume
Date
2004-11-18  44.38  44.49  44.07  44.43   5992000
2004-11-19  44.78  44.92  44.47  44.49  11655300
2004-11-22  44.95  44.97  44.74  44.75  11996000
2004-11-23  44.75  44.92  44.72  44.88   3169200
2004-11-24  45.05  45.05  44.79  44.93   6105100
…           …     …     …     …       …
2026-01-21 443.60 448.00 437.11 446.87  38830200
2026-01-22 451.79 452.98 443.56 443.84  19251200
2026-01-23 458.00 458.75 453.45 454.11  21531900
2026-01-26 464.70 469.28 460.36 466.78  34156000
2026-01-27 476.10 476.49 463.95 465.07  28386200

[5330 rows x 5 columns]
```

[40]:

```
gld = load_data(
    base_directory=DATA_DIR,
    ticker="GLD",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    timeframe="Daily",
    file_format="pickle",
)

# Filter GLD to date range
gld = gld[(gld.index >= pd.to_datetime(start_date)) & (gld.index <= pd.
 ↪to_datetime(end_date))]

# Resample to monthly frequency
gld_monthly = gld.resample("M").last()
gld_monthly["Monthly_Return"] = gld_monthly["Close"].pct_change()
```

[41]:

```
df_info(gld_monthly)
```

```
The columns, shape, and data types are:
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 252 entries, 2004-11-30 to 2025-10-31
Freq: ME
Data columns (total 6 columns):
```

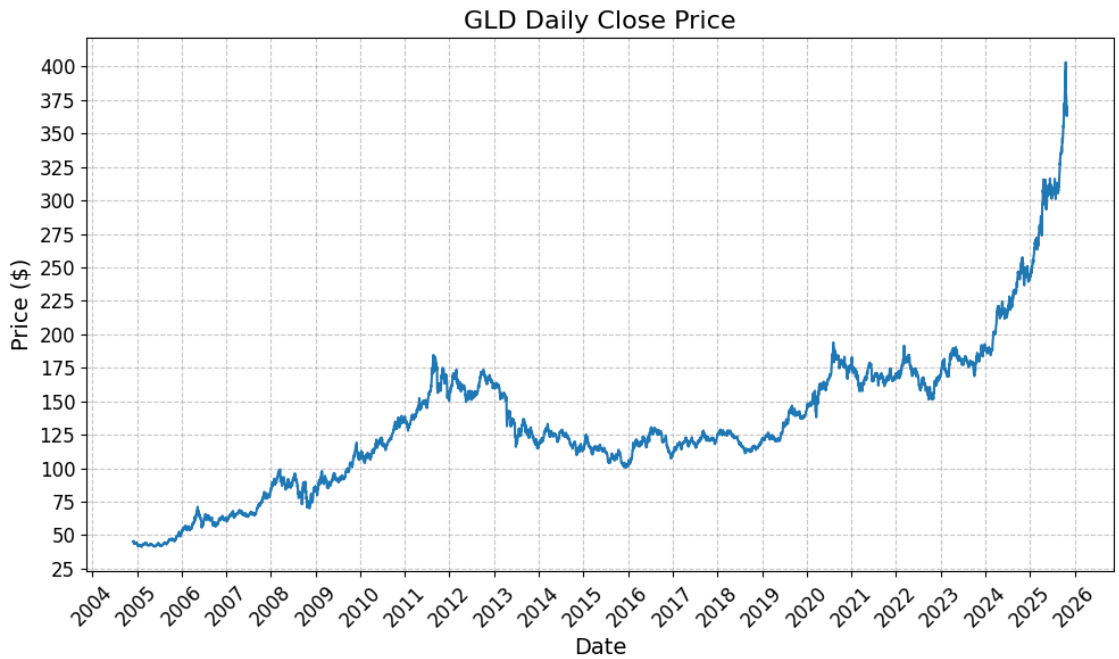
```
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Close           252 non-null    float64
 1   High            252 non-null    float64
 2   Low             252 non-null    float64
 3   Open            252 non-null    float64
 4   Volume          252 non-null    int64
 5   Monthly_Return  251 non-null    float64
dtypes: float64(5), int64(1)
memory usage: 13.8 KB
None
The first 5 rows are:

            Close  High   Low  Open   Volume  Monthly_Return
Date
2004-11-30  45.12 45.41 44.82 45.37  3857200             NaN
2004-12-31  43.80 43.94 43.73 43.85   531600           -0.03
2005-01-31  42.22 42.30 41.96 42.21  1692400           -0.04
2005-02-28  43.53 43.74 43.52 43.68   755300            0.03
2005-03-31  42.82 42.87 42.70 42.87  1363200           -0.02

The last 5 rows are:

            Close   High    Low   Open    Volume  Monthly_Return
Date
2025-06-30 304.83 304.92 301.95 302.39   8192100            0.00
2025-07-31 302.96 304.61 302.86 304.59   8981000           -0.01
2025-08-31 318.07 318.09 314.64 314.72  15642600            0.05
2025-09-30 355.47 355.57 350.87 351.13  13312400            0.12
2025-10-31 368.12 370.66 365.50 370.47  11077900            0.04
```

[42]:
```
# Copy this <!-- INSERT_04_GLD_Monthly_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="04_GLD_Monthly.md",
    content=df_info_markdown(df=gld_monthly, decimal_places=2),
    output_type="markdown",
)
```

```
Exported and tracked: 04_GLD_Monthly.md
```

[43]:
```
plot_timeseries(
    price_df=gld,
    plot_start_date=start_date,
    plot_end_date=end_date,
    plot_columns=["Close"],
    title="GLD Daily Close Price",
    x_label="Date",
    x_format="Year",
```

```
    x_tick_rotation=45,
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=25,
    grid=True,
    legend=False,
    export_plot=True,
    plot_file_name="04_GLD_Price",
)
```



[44]:
```
gld_cycle_df = calc_fed_cycle_asset_performance(
    fed_cycles=fed_cycles,
    cycle_labels=cycle_labels,
    fed_changes=fed_changes,
    monthly_df=gld_monthly,
)

gld_cycle_df
```

[44]:

	Cycle	Start	End	Months	CumulativeReturn	\
0	Cycle 1	2004-11-01	2006-07-01	20	0.36	
1	Cycle 2	2006-07-01	2007-07-01	12	0.05	
2	Cycle 3	2007-07-01	2008-12-01	17	0.25	
3	Cycle 4	2008-12-01	2015-11-01	83	0.36	

4	Cycle 5	2015-11-01	2019-01-01	38	0.11
5	Cycle 6	2019-01-01	2019-07-01	6	0.10
6	Cycle 7	2019-07-01	2020-04-01	9	0.11
7	Cycle 8	2020-04-01	2022-02-01	22	0.14
8	Cycle 9	2022-02-01	2023-08-01	18	0.08
9	Cycle 10	2023-08-01	2024-08-01	12	0.24
10	Cycle 11	2024-08-01	2026-01-28	15	0.62

	CumulativeReturnPct	AverageMonthlyReturn	AverageMonthlyReturnPct	\
0	35.70	0.02	1.73	
1	4.96	0.00	0.45	
2	24.96	0.02	1.59	
3	36.10	0.01	0.51	
4	10.93	0.00	0.35	
5	9.86	0.02	1.63	
6	11.15	0.01	1.24	
7	13.54	0.01	0.69	
8	8.48	0.01	0.53	
9	24.24	0.02	1.89	
10	62.49	0.03	3.36	

	AnnualizedReturn	AnnualizedReturnPct	Volatility	FedFundsChange	\
0	0.20	20.10	0.17	0.03	
1	0.05	4.96	0.11	0.00	
2	0.17	17.03	0.26	-0.05	
3	0.05	4.56	0.18	-0.00	
4	0.03	3.33	0.14	0.02	
5	0.21	20.68	0.12	0.00	
6	0.15	15.13	0.13	-0.02	
7	0.07	7.17	0.16	0.00	
8	0.06	5.58	0.14	0.05	
9	0.24	24.24	0.13	0.00	
10	0.47	47.46	0.14	-0.02	

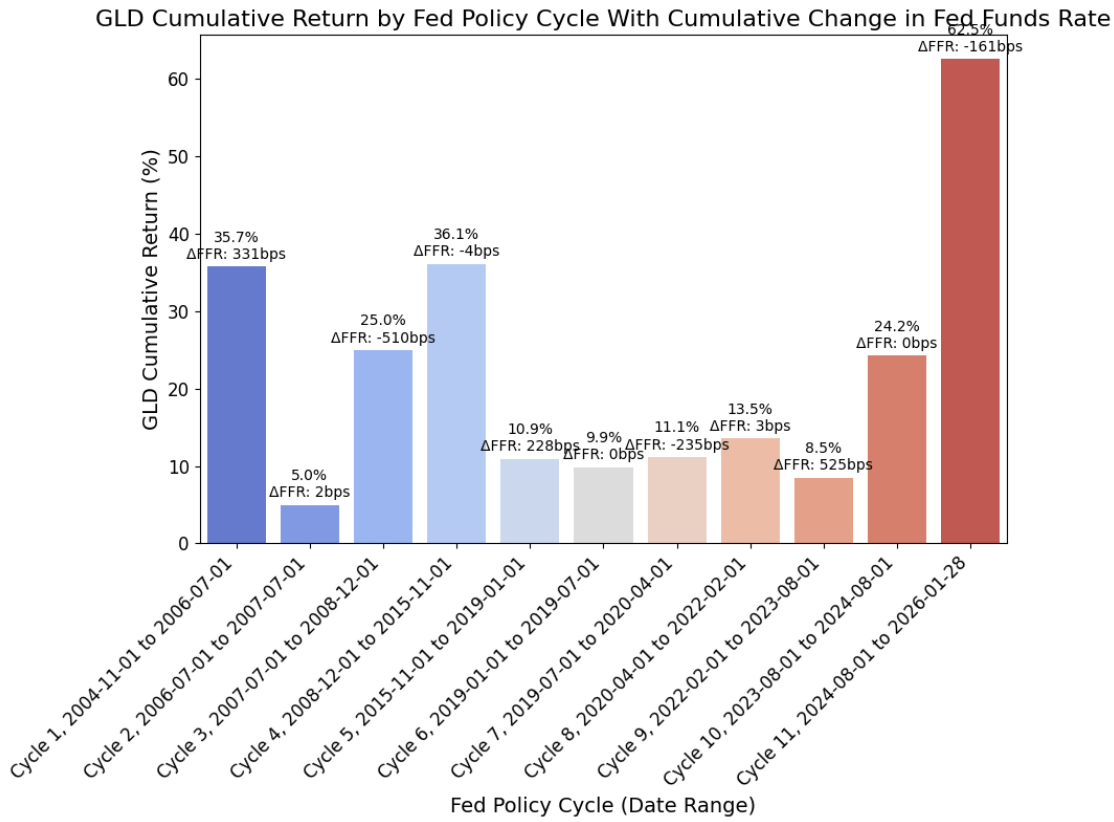
	FedFundsChange_bps	FFR_AnnualizedChange	FFR_AnnualizedChange_bps	\
0	331.00	0.02	198.60	
1	2.00	0.00	2.00	
2	-510.00	-0.04	-360.00	
3	-4.00	-0.00	-0.58	
4	228.00	0.01	72.00	
5	0.00	0.00	0.00	
6	-235.00	-0.03	-313.33	
7	3.00	0.00	1.64	
8	525.00	0.03	350.00	
9	0.00	0.00	0.00	
10	-161.00	-0.01	-128.80	

```
                                  Label
0    Cycle 1, 2004-11-01 to 2006-07-01
1    Cycle 2, 2006-07-01 to 2007-07-01
2    Cycle 3, 2007-07-01 to 2008-12-01
3    Cycle 4, 2008-12-01 to 2015-11-01
4    Cycle 5, 2015-11-01 to 2019-01-01
5    Cycle 6, 2019-01-01 to 2019-07-01
6    Cycle 7, 2019-07-01 to 2020-04-01
7    Cycle 8, 2020-04-01 to 2022-02-01
8    Cycle 9, 2022-02-01 to 2023-08-01
9   Cycle 10, 2023-08-01 to 2024-08-01
10  Cycle 11, 2024-08-01 to 2026-01-28
```

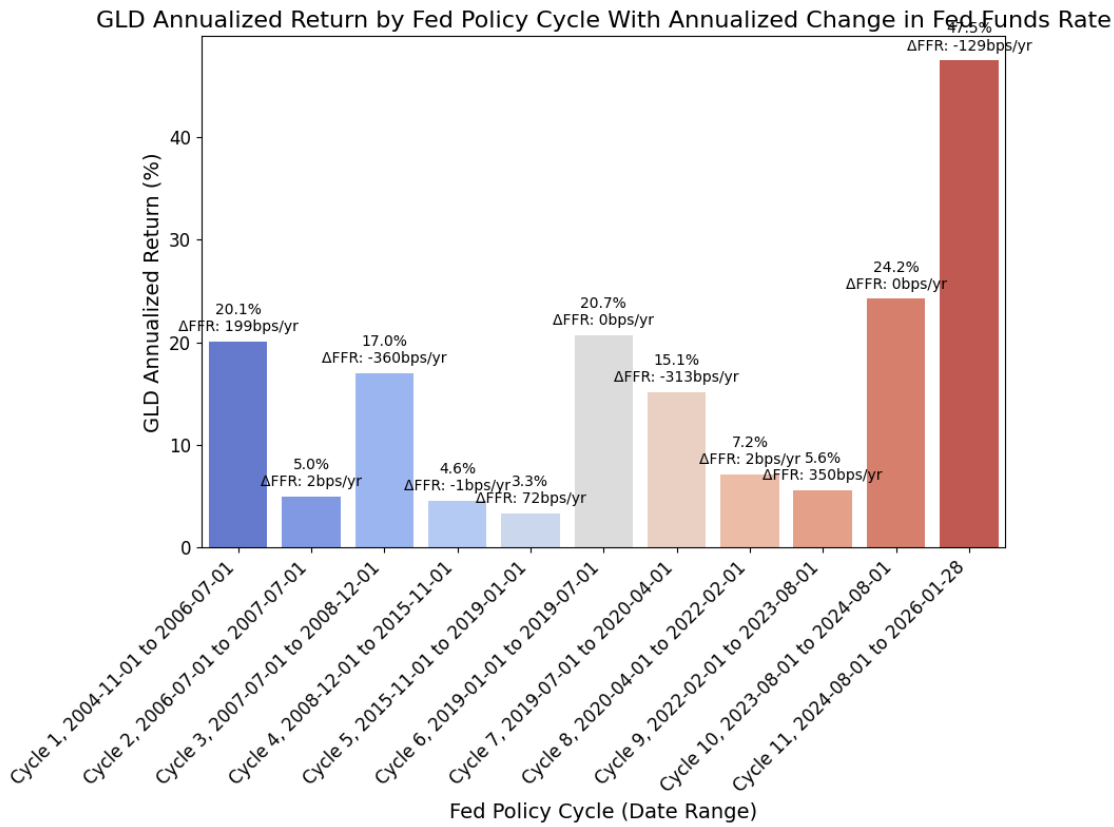
[45]:
```
# Copy this <!-- INSERT_04_GLD_Cycle_DF_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="04_GLD_Cycle_DF.md",
    content=gld_cycle_df.to_markdown(floatfmt=".2f"),
    output_type="markdown",
)
```

```
Exported and tracked: 04_GLD_Cycle_DF.md
```

[46]:
```
plot_bar_returns_ffr_change(
    cycle_df=gld_cycle_df,
    asset_label="GLD",
    annualized_or_cumulative="Cumulative",
    index_num="04",
)
```



[47]:
```
plot_bar_returns_ffr_change(
    cycle_df=gld_cycle_df,
    asset_label="GLD",
    annualized_or_cumulative="Annualized",
    index_num="04",
)
```



[48]:
```
df = gld_cycle_df

#####################################
### Don't modify below this line ###
#####################################

# Run OLS regression with statsmodels
X = df["FFR_AnnualizedChange_bps"]
y = df["AnnualizedReturnPct"]
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())
print(f"Intercept: {model.params[0]}, Slope: {model.params[1]}")  # Intercept␣
↪and slope

# Calc X and Y values for regression line
X_vals = np.linspace(X.min(), X.max(), 100)
Y_vals = model.params[0] + model.params[1] * X_vals
```

```
                            OLS Regression Results
==============================================================================
```

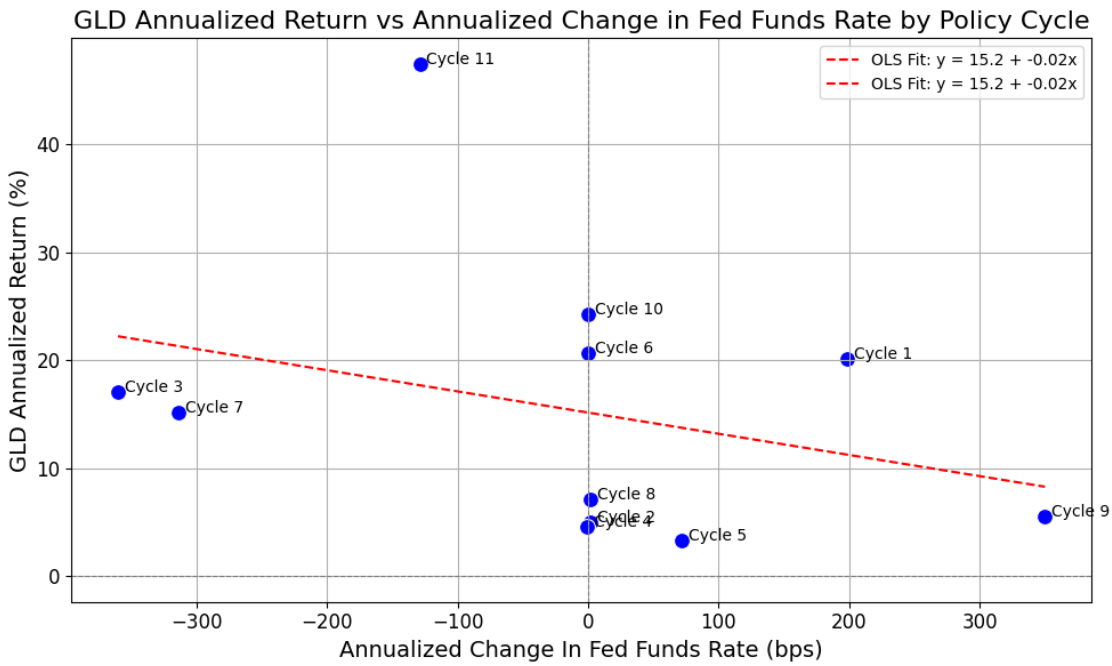
```
Dep. Variable:     AnnualizedReturnPct   R-squared:                       0.093
Model:                             OLS   Adj. R-squared:                 -0.008
Method:                  Least Squares   F-statistic:                    0.9214
Date:                 Wed, 28 Jan 2026   Prob (F-statistic):              0.362
Time:                         19:50:08   Log-Likelihood:                -42.778
No. Observations:                   11   AIC:                             89.56
Df Residuals:                        9   BIC:                             90.35
Df Model:                            1
Covariance Type:             nonrobust
==============================================================================
============
                              coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
------------
const                       15.1586      3.955      3.833      0.004       6.213
24.104
FFR_AnnualizedChange_bps    -0.0196      0.020     -0.960      0.362      -0.066
0.027
==============================================================================
Omnibus:                        7.682   Durbin-Watson:                   0.913
Prob(Omnibus):                  0.021   Jarque-Bera (JB):                3.504
Skew:                           1.305   Prob(JB):                        0.173
Kurtosis:                       3.912   Cond. No.                         194.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
Intercept: 15.158628269088016, Slope: -0.019626313878624208
```

[49]:
```
# Copy this <!-- INSERT_04_GLD_Annualized_Regression_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="04_GLD_Annualized_Regression.md",
    content=sm_ols_summary_markdown(result=model,␣
 ↪file_path="04_GLD_Annualized_Regression.md"),
    output_type="text",
)
```

```
Exported and tracked: 04_GLD_Annualized_Regression.md
```

[50]:
```
plot_scatter_regression_ffr_vs_returns(
    cycle_df=gld_cycle_df,
    asset_label="GLD",
    index_num="04",
    x_vals=X_vals,
    y_vals=Y_vals,
```

```
    intercept=model.params[0],
    slope=model.params[1],
)
```



1.7 Hybrid Portfolio

1.7.1 Asset Allocation

[51]:
```
fed_cycles
```

[51]:
```
[('2004-11-01', '2006-07-01'),
 ('2006-07-01', '2007-07-01'),
 ('2007-07-01', '2008-12-01'),
 ('2008-12-01', '2015-11-01'),
 ('2015-11-01', '2019-01-01'),
 ('2019-01-01', '2019-07-01'),
 ('2019-07-01', '2020-04-01'),
 ('2020-04-01', '2022-02-01'),
 ('2022-02-01', '2023-08-01'),
 ('2023-08-01', '2024-08-01'),
 ('2024-08-01', '2026-01-28')]
```

[52]:
```
cycle_labels
```

[52]:
```
['Cycle 1',
 'Cycle 2',
```

```
 'Cycle 3',
 'Cycle 4',
 'Cycle 5',
 'Cycle 6',
 'Cycle 7',
 'Cycle 8',
 'Cycle 9',
 'Cycle 10',
 'Cycle 11']
```

[53]:
```
# Calculate cumulative returns and drawdown for SPY
spy_monthly['Cumulative_Return'] = (1 + spy_monthly['Monthly_Return']).
 ↪cumprod() - 1
spy_monthly['Cumulative_Return_Plus_One'] = 1 + spy_monthly['Cumulative_Return']
spy_monthly['Rolling_Max'] = spy_monthly['Cumulative_Return_Plus_One'].cummax()
spy_monthly['Drawdown'] = spy_monthly['Cumulative_Return_Plus_One'] /␣
 ↪spy_monthly['Rolling_Max'] - 1
spy_monthly.drop(columns=['Cumulative_Return_Plus_One', 'Rolling_Max'],␣
 ↪inplace=True)

# Calculate cumulative returns and drawdown for TLT
tlt_monthly['Cumulative_Return'] = (1 + tlt_monthly['Monthly_Return']).
 ↪cumprod() - 1
tlt_monthly['Cumulative_Return_Plus_One'] = 1 + tlt_monthly['Cumulative_Return']
tlt_monthly['Rolling_Max'] = tlt_monthly['Cumulative_Return_Plus_One'].cummax()
tlt_monthly['Drawdown'] = tlt_monthly['Cumulative_Return_Plus_One'] /␣
 ↪tlt_monthly['Rolling_Max'] - 1
tlt_monthly.drop(columns=['Cumulative_Return_Plus_One', 'Rolling_Max'],␣
 ↪inplace=True)

# Isolate the returns for SPY and TLT
spy_ret = spy_monthly['Monthly_Return']
tlt_ret = tlt_monthly['Monthly_Return']

# Create a blended portfolio based on Fed policy cycles
portfolio = (
    spy_ret[spy_ret.index <= "2007-07-01"]
    .combine_first(tlt_ret[(tlt_ret.index >= "2007-07-01") & (tlt_ret.index <=␣
 ↪"2008-12-01")])
    .combine_first(spy_ret[(spy_ret.index > "2008-12-01") & (spy_ret.index <=␣
 ↪"2019-07-01")])
    .combine_first(tlt_ret[(tlt_ret.index >= "2019-07-01") & (tlt_ret.index <=␣
 ↪"2020-04-01")])
    .combine_first(spy_ret[(spy_ret.index > "2020-04-01") & (spy_ret.index <=␣
 ↪"2024-08-01")])
    .combine_first(tlt_ret[tlt_ret.index > "2024-08-01"])
```

```
)

# Convert to DataFrame
portfolio_monthly = portfolio.to_frame(name="Portfolio_Monthly_Return")

# Calculate cumulative returns and drawdown for the portfolio
portfolio_monthly['Portfolio_Cumulative_Return'] = (1 +␣
 ↪portfolio_monthly['Portfolio_Monthly_Return']).cumprod() - 1
portfolio_monthly['Portfolio_Cumulative_Return_Plus_One'] = 1 +␣
 ↪portfolio_monthly['Portfolio_Cumulative_Return']
portfolio_monthly['Portfolio_Rolling_Max'] =␣
 ↪portfolio_monthly['Portfolio_Cumulative_Return_Plus_One'].cummax()
portfolio_monthly['Portfolio_Drawdown'] =␣
 ↪portfolio_monthly['Portfolio_Cumulative_Return_Plus_One'] /␣
 ↪portfolio_monthly['Portfolio_Rolling_Max'] - 1
portfolio_monthly.drop(columns=['Portfolio_Cumulative_Return_Plus_One',␣
 ↪'Portfolio_Rolling_Max'], inplace=True)

# Merge "spy_monthly" and "tlt_monthly" into "portfolio_monthly" to compare␣
 ↪cumulative returns
portfolio_monthly = portfolio_monthly.join(
    spy_monthly['Monthly_Return'].rename('SPY_Monthly_Return'),
    how='left'
).join(
    spy_monthly['Cumulative_Return'].rename('SPY_Cumulative_Return'),
    how='left'
).join(
    spy_monthly['Drawdown'].rename('SPY_Drawdown'),
    how='left'
).join(
    tlt_monthly['Monthly_Return'].rename('TLT_Monthly_Return'),
    how='left'
).join(
    tlt_monthly['Cumulative_Return'].rename('TLT_Cumulative_Return'),
    how='left'
).join(
    tlt_monthly['Drawdown'].rename('TLT_Drawdown'),
    how='left'
)
```

[54]:
```
df_info(portfolio_monthly)
```

```
The columns, shape, and data types are:
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 252 entries, 2004-11-30 to 2025-10-31
Freq: ME
Data columns (total 9 columns):
```

```
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Portfolio_Monthly_Return     251 non-null    float64
 1   Portfolio_Cumulative_Return  251 non-null    float64
 2   Portfolio_Drawdown           251 non-null    float64
 3   SPY_Monthly_Return           251 non-null    float64
 4   SPY_Cumulative_Return        251 non-null    float64
 5   SPY_Drawdown                 251 non-null    float64
 6   TLT_Monthly_Return           251 non-null    float64
 7   TLT_Cumulative_Return        251 non-null    float64
 8   TLT_Drawdown                 251 non-null    float64
dtypes: float64(9)
memory usage: 19.7 KB
None
The first 5 rows are:

            Portfolio_Monthly_Return  Portfolio_Cumulative_Return  \
Date
2004-11-30                       NaN                          NaN
2004-12-31                      0.03                         0.03
2005-01-31                     -0.02                         0.01
2005-02-28                      0.02                         0.03
2005-03-31                     -0.02                         0.01

            Portfolio_Drawdown  SPY_Monthly_Return  SPY_Cumulative_Return  \
Date
2004-11-30                 NaN                 NaN                    NaN
2004-12-31                0.00                0.03                   0.03
2005-01-31               -0.02               -0.02                   0.01
2005-02-28               -0.00                0.02                   0.03
2005-03-31               -0.02               -0.02                   0.01

            SPY_Drawdown  TLT_Monthly_Return  TLT_Cumulative_Return  \
Date
2004-11-30           NaN                 NaN                    NaN
2004-12-31          0.00                0.03                   0.03
2005-01-31         -0.02                0.04                   0.06
2005-02-28         -0.00               -0.01                   0.05
2005-03-31         -0.02               -0.00                   0.04

            TLT_Drawdown
Date
2004-11-30           NaN
2004-12-31          0.00
2005-01-31          0.00
2005-02-28         -0.01
2005-03-31         -0.02

The last 5 rows are:
```

```
            Portfolio_Monthly_Return  Portfolio_Cumulative_Return  \
Date
2025-06-30                      0.03                        19.00
2025-07-31                     -0.01                        18.78
2025-08-31                      0.00                        18.78
2025-09-30                      0.04                        19.49
2025-10-31                      0.01                        19.77

            Portfolio_Drawdown  SPY_Monthly_Return  SPY_Cumulative_Return  \
Date
2025-06-30               -0.07                0.05                   6.72
2025-07-31               -0.08                0.02                   6.90
2025-08-31               -0.08                0.02                   7.06
2025-09-30               -0.05                0.04                   7.34
2025-10-31               -0.04                0.02                   7.54

            SPY_Drawdown  TLT_Monthly_Return  TLT_Cumulative_Return  \
Date
2025-06-30          0.00                0.03                   0.96
2025-07-31          0.00               -0.01                   0.94
2025-08-31          0.00                0.00                   0.94
2025-09-30          0.00                0.04                   1.01
2025-10-31          0.00                0.01                   1.04

            TLT_Drawdown
Date
2025-06-30         -0.41
2025-07-31         -0.41
2025-08-31         -0.41
2025-09-30         -0.39
2025-10-31         -0.39
```

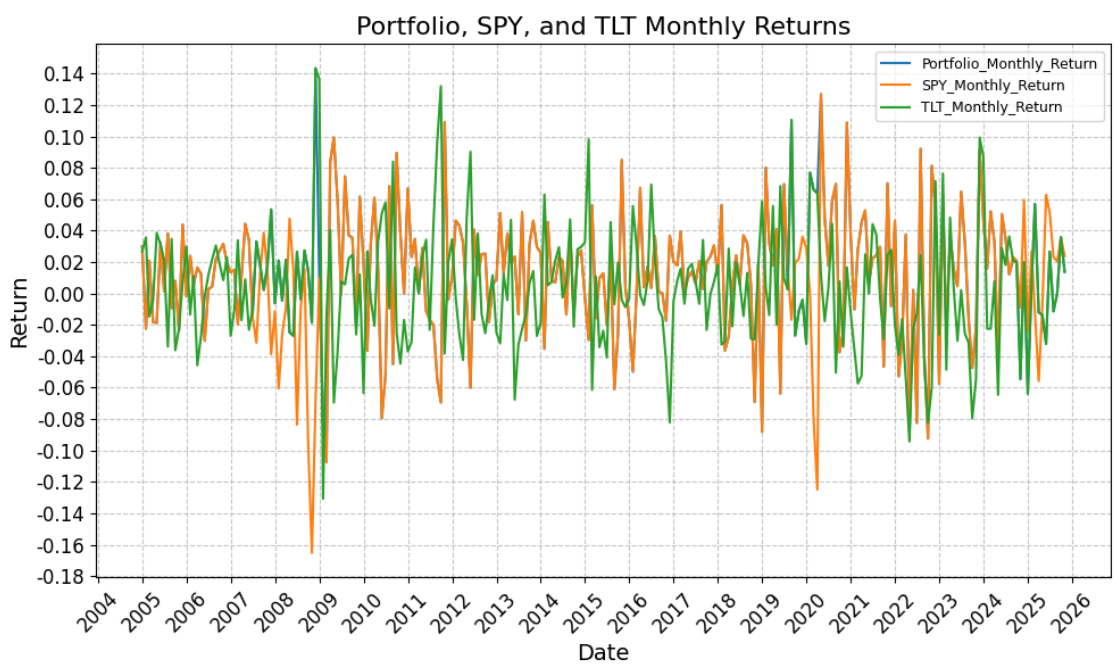
[55]:
```
# Copy this <!-- INSERT_05_Portfolio_DF_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="05_Portfolio_DF.md",
    content=df_info_markdown(df=portfolio_monthly, decimal_places=3),
    output_type="markdown",
)
```

```
Exported and tracked: 05_Portfolio_DF.md
```

1.7.2 Performance Statistics

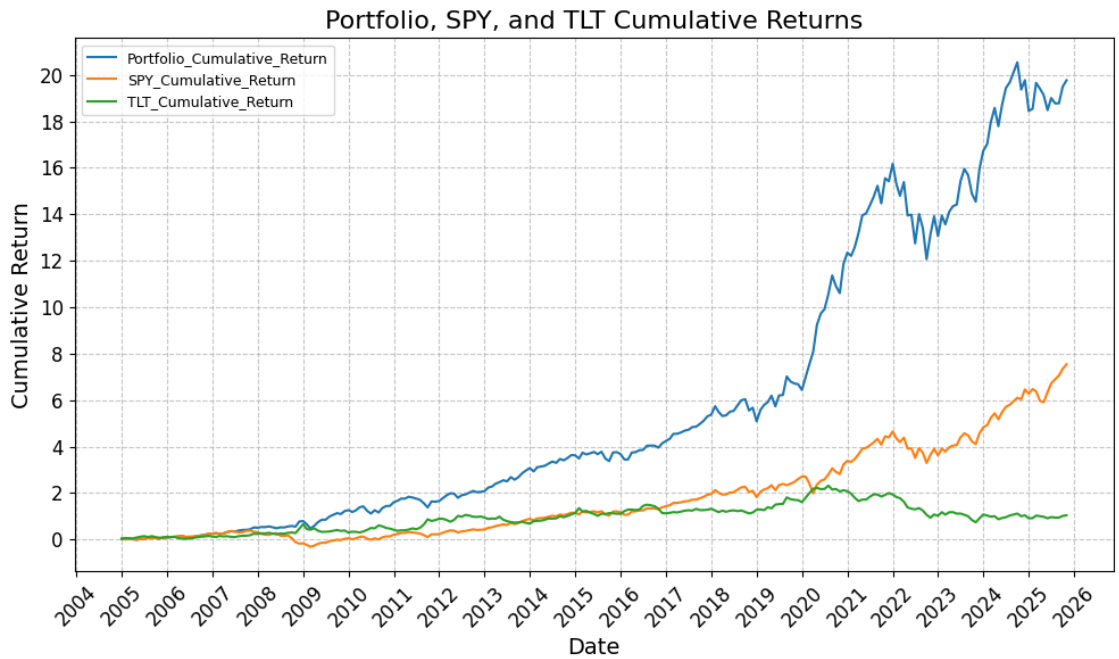
[56]:
```
plot_timeseries(
    price_df=portfolio_monthly,
    plot_start_date=start_date,
    plot_end_date=end_date,
```

```
    plot_columns=["Portfolio_Monthly_Return", "SPY_Monthly_Return",␣
↪"TLT_Monthly_Return"],
    title="Portfolio, SPY, and TLT Monthly Returns",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Return",
    y_format="Decimal",
    y_format_decimal_places=2,
    y_tick_spacing=0.02,
    grid=True,
    legend=True,
    export_plot=True,
    plot_file_name="05_Monthly_Returns",
)
```



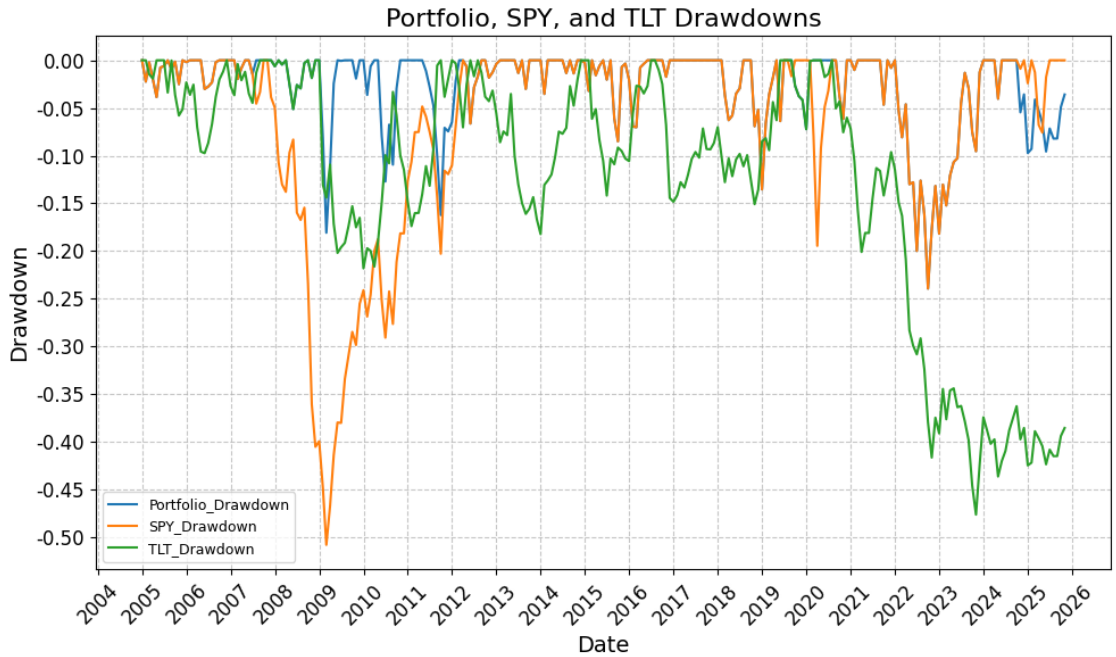
[57]:
```
plot_timeseries(
    price_df=portfolio_monthly,
    plot_start_date=start_date,
    plot_end_date=end_date,
    plot_columns=["Portfolio_Cumulative_Return", "SPY_Cumulative_Return",␣
↪"TLT_Cumulative_Return"],
    title="Portfolio, SPY, and TLT Cumulative Returns",
    x_label="Date",
    x_format="Year",
```

```
    x_tick_rotation=45,
    y_label="Cumulative Return",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=2,
    grid=True,
    legend=True,
    export_plot=True,
    plot_file_name="05_Cumulative_Returns",
)
```



[58]:
```
plot_timeseries(
    price_df=portfolio_monthly,
    plot_start_date=start_date,
    plot_end_date=end_date,
    plot_columns=["Portfolio_Drawdown", "SPY_Drawdown", "TLT_Drawdown"],
    title="Portfolio, SPY, and TLT Drawdowns",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Drawdown",
    y_format="Decimal",
    y_format_decimal_places=2,
    y_tick_spacing=0.05,
    grid=True,
```

```
    legend=True,
    export_plot=True,
    plot_file_name="05_Drawdowns",
)
```



[59]:
```
port_sum_stats = summary_stats(
    fund_list=["Portfolio", "SPY", "TLT"],
    df=portfolio_monthly[["Portfolio_Monthly_Return"]],
    period="Monthly",
    use_calendar_days=False,
    excel_export=False,
    pickle_export=False,
    output_confirmation=False,
)

spy_sum_stats = summary_stats(
    fund_list=["Portfolio", "SPY", "TLT"],
    df=portfolio_monthly[["SPY_Monthly_Return"]],
    period="Monthly",
    use_calendar_days=False,
    excel_export=False,
    pickle_export=False,
    output_confirmation=False,
)
```

```
tlt_sum_stats = summary_stats(
    fund_list=["Portfolio", "SPY", "TLT"],
    df=portfolio_monthly[["TLT_Monthly_Return"]],
    period="Monthly",
    use_calendar_days=False,
    excel_export=False,
    pickle_export=False,
    output_confirmation=False,
)

sum_stats = pd.concat([port_sum_stats, spy_sum_stats, tlt_sum_stats])
sum_stats
```

[59]:

```
                          Arithmetic Annual Mean Return  \
Portfolio_Monthly_Return                            0.16
SPY_Monthly_Return                                  0.11
TLT_Monthly_Return                                  0.04

                          Annualized Volatility  Annualized Sharpe Ratio  \
Portfolio_Monthly_Return                   0.14                      1.11
SPY_Monthly_Return                         0.15                      0.77
TLT_Monthly_Return                         0.14                      0.32

                          CAGR  Monthly Max Return Monthly Max Return (Date)  \
Portfolio_Monthly_Return  0.16                0.14                2008-11-30
SPY_Monthly_Return        0.11                0.13                2020-04-30
TLT_Monthly_Return        0.03                0.14                2008-11-30

                          Monthly Min Return Monthly Min Return (Date)  \
Portfolio_Monthly_Return               -0.11                2009-02-28
SPY_Monthly_Return                     -0.17                2008-10-31
TLT_Monthly_Return                     -0.13                2009-01-31

                          Max Drawdown       Peak     Trough Recovery Date  \
Portfolio_Monthly_Return         -0.24 2021-12-31 2022-09-30    2023-12-31
SPY_Monthly_Return               -0.51 2007-10-31 2009-02-28    2012-03-31
TLT_Monthly_Return               -0.48 2020-07-31 2023-10-31           NaT

                          Days to Recover  MAR Ratio
Portfolio_Monthly_Return           457.00       0.65
SPY_Monthly_Return                1127.00       0.21
TLT_Monthly_Return                    NaN       0.07
```

[60]:

```
# Copy this <!-- INSERT_05_Portfolio_Stats_DF_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="05_Portfolio_Stats_DF.md",
```

```
    content=sum_stats.to_markdown(floatfmt=".3f"),
    output_type="markdown",
)
```

```
Exported and tracked: 05_Portfolio_Stats_DF.md
```

[ ]:

```

```